FLAVORQUEST

AI-Powered Culinary Advisor Project Synopsis

Prepared by: Girish Jangid (gj6253@my.utexas.edu)

Course / Team / Institution: Android Programming, University Of Texas, Austin

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App Title

FlavorQuest

Group Members & Contact Emails

• Girish Jangid

Project Overview

FlavorQuest is an AI-powered culinary companion designed to simplify everyday food decisions by understanding each user's unique preferences, mood, and lifestyle. Whether someone feels like dining out or preparing a home-cooked meal, the app intelligently guides them with personalized suggestions tailored to dietary needs, time constraints, and taste preferences. By integrating generative AI with real-time location services, FlavorQuest can recommend nearby restaurants along with curated menu highlights or provide easy-to-follow recipes using ingredients users already have at home.

The application creates a seamless experience by learning from user interactions, improving its recommendations over time to feel more intuitive and human-like. It also considers factors such as cuisine preferences, health goals, and even weather or time of day to enhance its suggestions. With a clean and user-friendly interface, FlavorQuest aims to reduce decision fatigue and make meal planning enjoyable. Ultimately, it serves as a smart food assistant that bridges the gap between convenience and creativity in everyday dining choices.

Major Subsystems

• User Authentication & Profile Management

Handles secure user registration, login, and profile customization, including dietary preferences, allergies, favorite cuisines, and past activity to enable personalized experiences.

• User Input Processing & Decision Engine

Captures user intent such as mood, time availability, and choice between cooking or ordering, and processes this data to determine the most relevant type of recommendation.

• AI Interaction & Integration Layer

Connects the application with AI services to generate intelligent responses, recipe suggestions, and contextual recommendations through a conversational interface.

• Recommendation & Location Services Module

Delivers tailored outputs including nearby restaurant options using real-time location data, as well as curated recipes with ingredients, steps, and preparation details.

Major Technical Challenges

• Prompt Engineering & JSON Parsing

Designing precise prompts to generate structured, reliable AI responses and safely parsing them into usable application data.

• State Management for Conversational Refinement

Maintaining context across user interactions to ensure continuity and progressively improved recommendations.

• Asynchronous Network Handling

Managing multiple API calls (AI, restaurant data, maps) efficiently without blocking the user interface or degrading performance.

• Location Accuracy & Real-Time Geolocation Handling

Ensuring precise user location detection, handling permissions, and dynamically updating nearby restaurant recommendations based on movement and availability.

• Personalization & Recommendation Accuracy

Balancing AI-generated suggestions with user preferences, history, and constraints to deliver relevant and trustworthy results.

• API Integration & Data Consistency

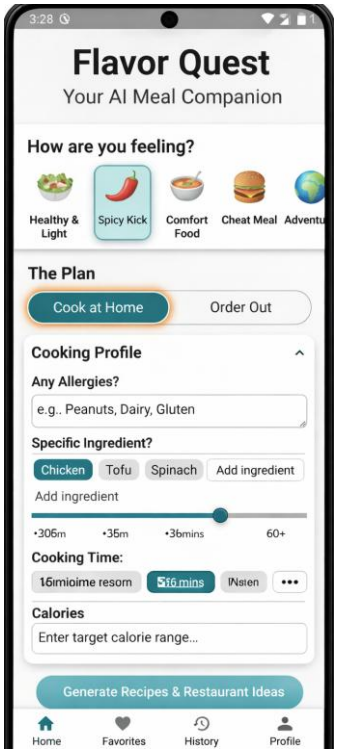
Coordinating data from multiple third-party services while handling failures, inconsistencies, and rate limits.

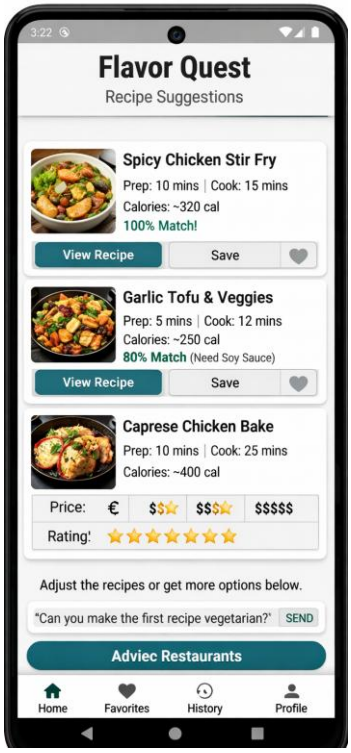
• Error Handling & Fallback Mechanisms

Providing graceful degradation when AI responses fail, APIs are unavailable, or incomplete data is returned.

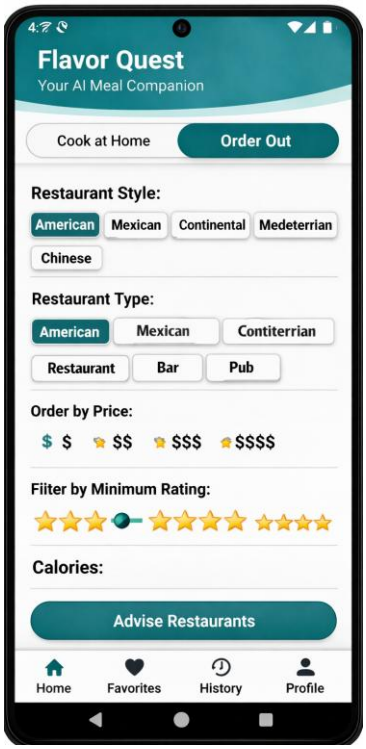
User Interface Sketches

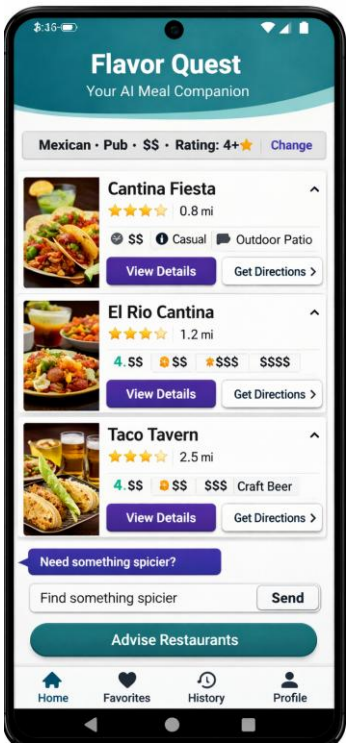
The following mockups illustrate the major UI components of the FlavorQuest app:

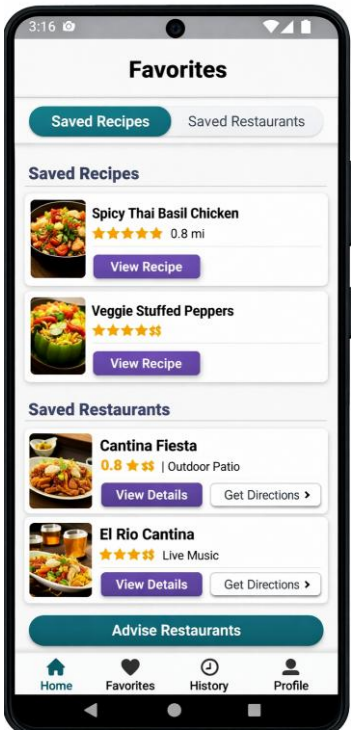


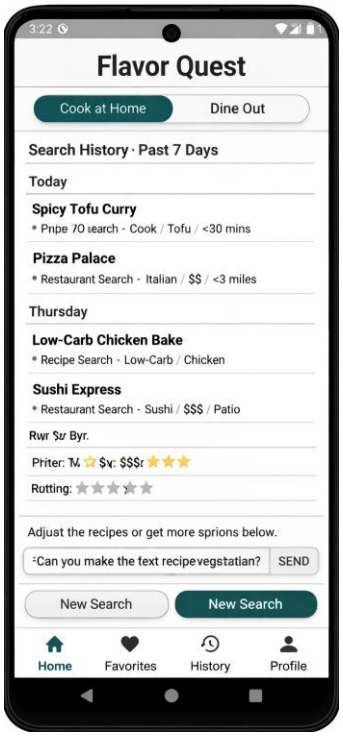


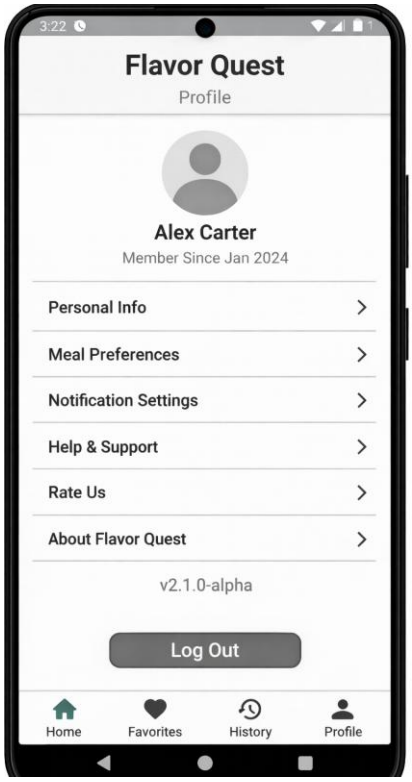
Ordering Out:






3:16
Favorites
Saved Recipes
Saved Restaurants
Saved Recipes
Spicy Thai Basil Chicken
0.8 mi
View Recipe
Veggie Stuffed Peppers
View Recipe
Saved Restaurants
Cantina Fiesta
0.8 $$ | Outdoor Patio
View Details
Get Directions
El Rio Cantina
$$$ Live Music
View Details
Get Directions
Advise Restaurants
Home
Favorites
History
Profile


3:22
Flavor Quest
Cook at Home
Dine Out
Search History · Past 7 Days
Today
Spicy Tofu Curry
• Pnpe 70 search - Cook / Tofu / <30 mins
Pizza Palace
• Restaurant Search - Italian / $$ / <3 miles
Thursday
Low-Carb Chicken Bake
• Recipe Search - Low-Carb / Chicken
Sushi Express
• Restaurant Search - Sushi / $$$ / Patio
Rwr $ur Byr.
Priter: 7M $x: $$$r
Rotting:
Adjust the recipes or get more sprions below.
:Can you make the fext recipe vegstatian?
SEND
New Search
New Search
Home
Favorites
History
Profile


3:22
Flavor Quest
Profile
Alex Carter
Member Since Jan 2024
Personal Info
Meal Preferences
Notification Settings
Help & Support
Rate Us
About Flavor Quest
v2.1.0-alpha
Log Out
Home
Favorites
History
Profile

Week-by-Week Schedule

Week	Focus Area	Deliverables
1	Project Setup & Auth	Android Studio setup, Firebase Auth, navigation
2	UI & Input Flow	Mood selection, Cook/Order branching
3	AI Integration	REST client, prompt logic, async handling
4	Parsing, Testing & Demo Prep	Recipe parsing, Google Places integration, usability polish

APIs & Libraries

- Llama Inference API (HuggingFace / Replicate / Groq)
- Google Places API & Maps SDK
- Firebase Authentication
- Retrofit + Gson/Moshi
- Firebase Firestore

Functional Specification

- User logs in via Firebase Auth
- Welcome screen: mood, allergies, cook/order toggle
- Cook Mode: ingredients, time, cuisine, flavor
- Order Out Mode: search radius
- AI prompt construction and query
- Display recipe or restaurant cards
- Chat refinement
- Save to local database
- Start Over resets session

Similar Apps & Inspiration

- Supercook / MyFridgeFood
- Yelp / TripAdvisor

Technical Scope, Usability & Demo Plan

Core Functional Areas

- Third-party LLM integration
- Location-aware restaurant matching
- Dynamic, context-aware UI

Usability & Reset Flow

- Dedicated Start Over button clears ViewModel state and resets navigation

Demo Preparation

- Pre-seeded recipe Vault
- Mocked location data
- Live Llama API calls